

HW4

1) a) Let:

$X_1 = \# \text{ vests}$
 $X_2 = \# \text{ lifeboats}$

LP:

Max. X_2

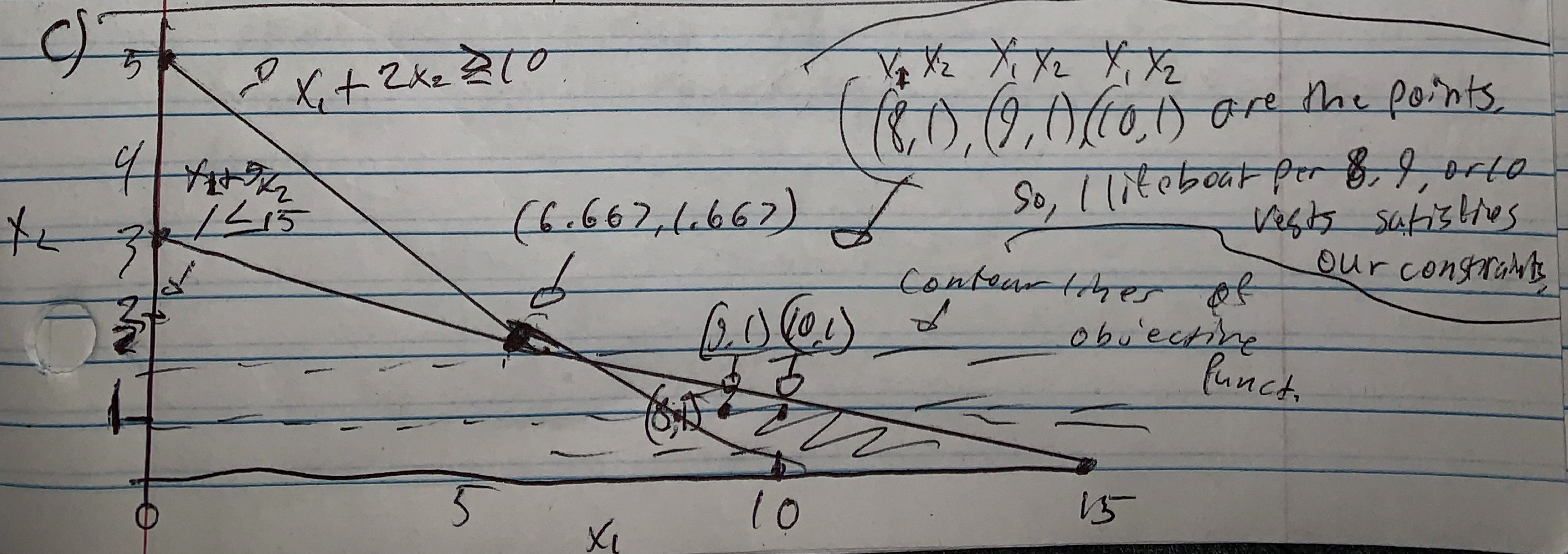
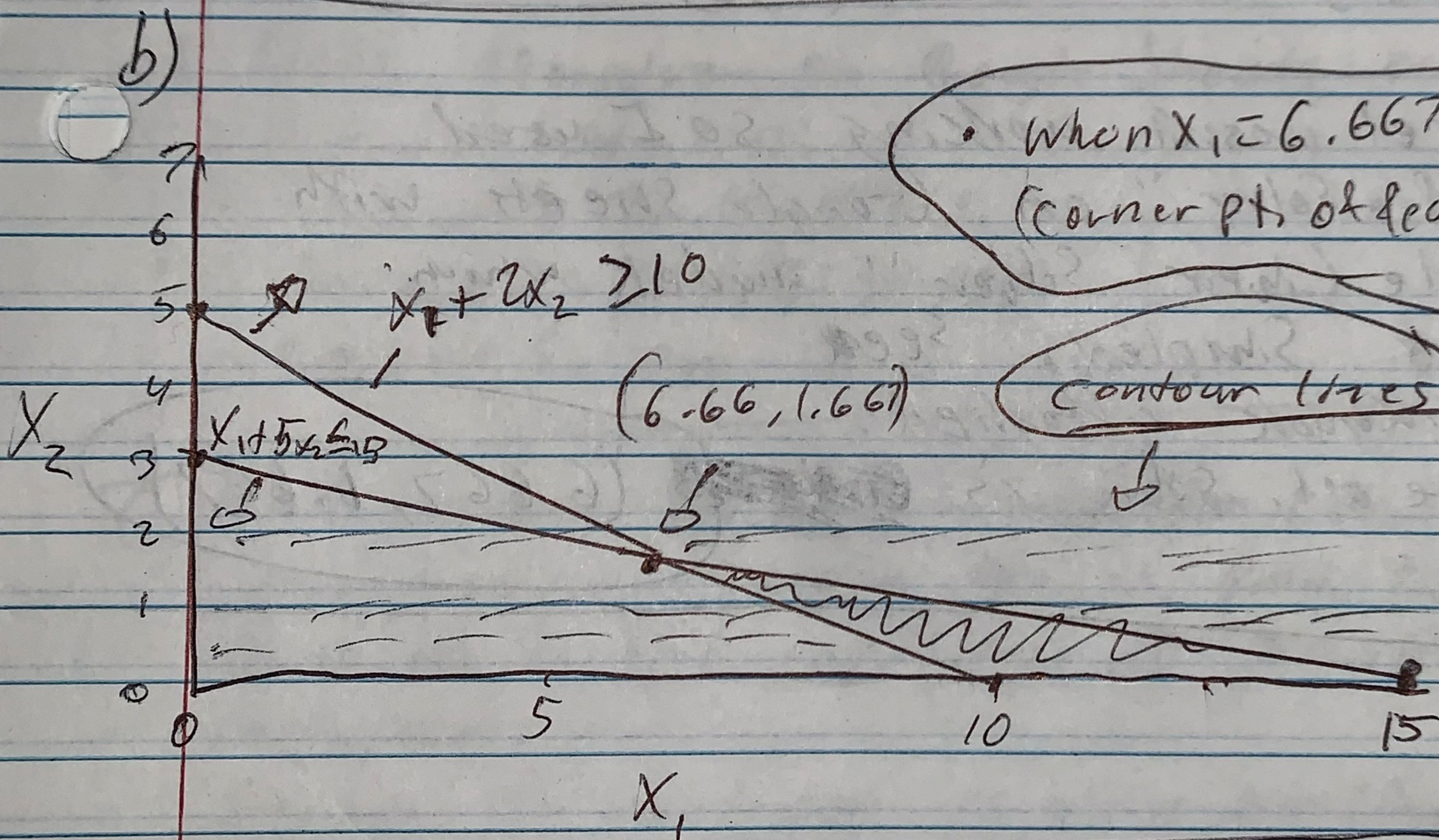
s.t.,

$$1x_1 + 2x_2 \geq 10$$

$$1x_1 + 5x_2 \leq 15$$

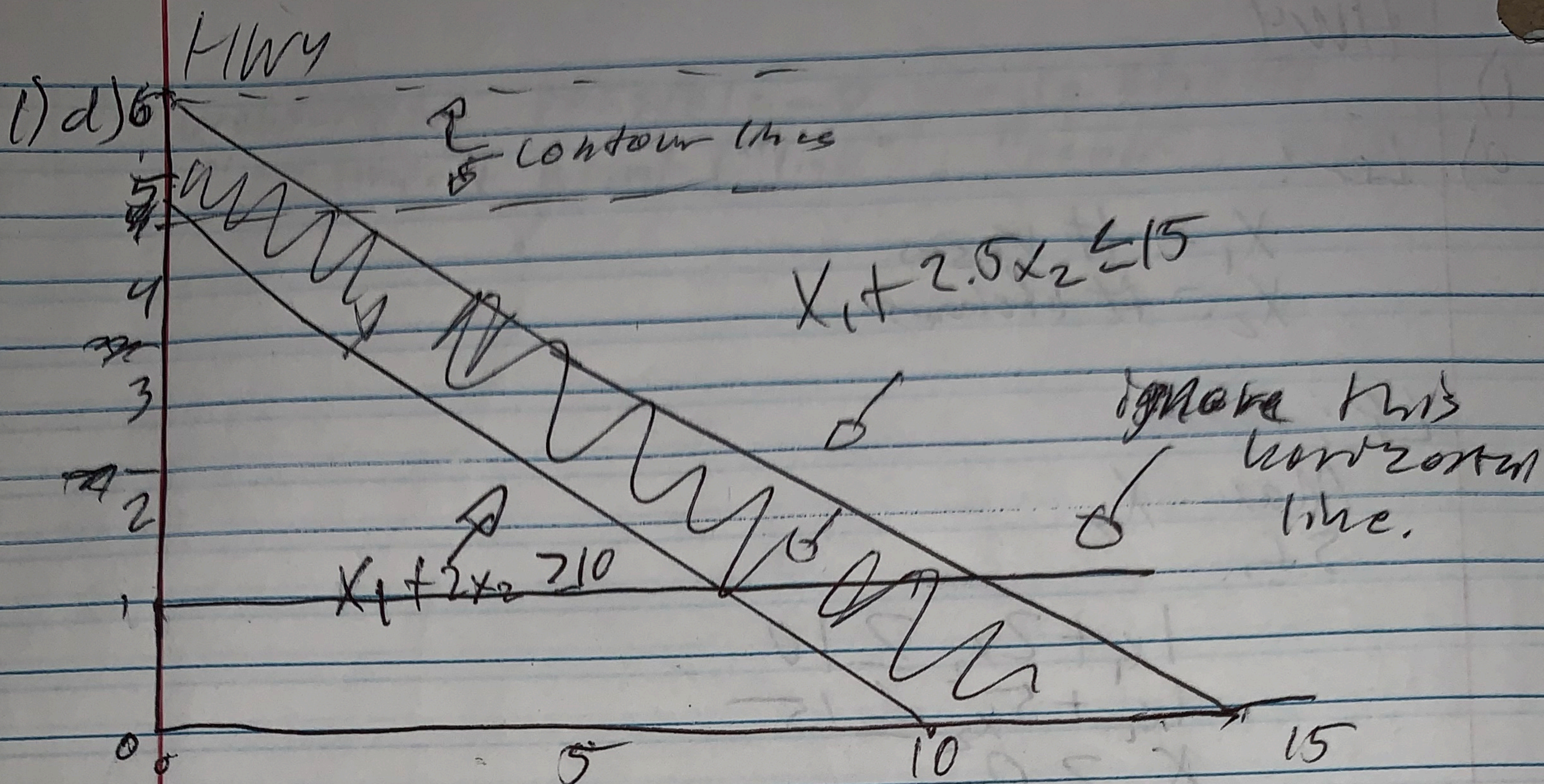
$$x_1 \geq 0$$

$$x_2 \geq 0$$



Mat 210

015-70M



with this updated graph, the corner pt. maximizing life boats is $(0, 6) \rightarrow 0$ vests, 6 boats.

c) Excel wasn't working, so I used "Open Solver" on Google Sheets with "Google Linear Solver" model which uses Simplex. See workbook attached.

The opt. soln. is ~~6.667, 1.667~~ $(6.667, 1.667)$

Mar 210

TIWY

2) Let:

- a) $a = \text{units (Product 1)}$
 $b = \text{units (Product 2)}$
 $c = \text{units (Product 3)}$

LP:

Max: $30a + 12b + 15c$

S.t.

$$9a + 3b + 5c \leq 500$$

$$5a + 4b \leq 350$$

$$3a + 2c \leq 150$$

$$c \leq 20$$

$$a, b, c \geq 0$$

b) Using QPensolver on Google Sheets, our optimal soln. is:

- (a) Product 1: 26,19047619
(b) Product 2: 54,76190476
(c) Product 3: 20

c) The mailing machine & ~~lathe~~ constraints are binding, because the opt. prod. plans uses their full capacities of 500 & 350, respectively.
Product 3's sales cap. is also a binding, reaching 20/20. The lathe isn't binding, only reaching ~118/150 possible hours.

Mat 210

MW 4

3a)

Let:

x_1 = hours worked / week

x_2 = class hours / wk

LP:

Maximize x_2 :

s.t.

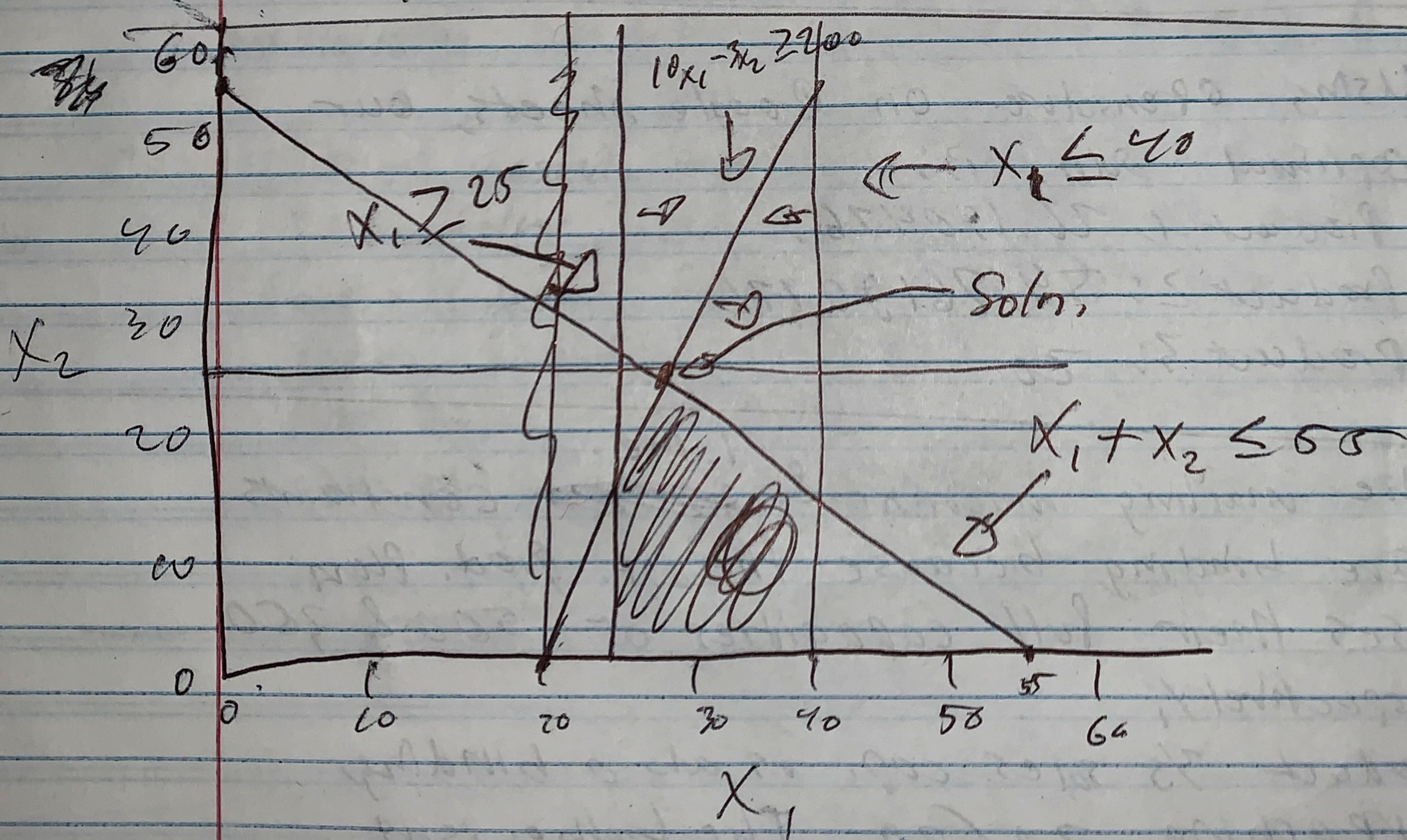
$$x_1 \geq 25$$

$$x_1 \leq 40$$

$$x_1 + x_2 \leq 55$$

$$10x_1 - 3x_2 \geq 200$$

$$x_1, x_2 \geq 0$$



- Soln maximizing class hrs. is corner pt. where constraints $x_1 + x_2 \leq 55$ & $10x_1 - 3x_2 \geq 200$ intersect - feasible region + class hrs are maxed.

MAT210

HW9

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Soln on intersection of $x_1 + x_2 \leq 55$ &

$$10x_1 - 3x_2 \geq 200$$

$$x_1 + x_2 = 55 \rightarrow x_1 = 55 - x_2$$

$$10x_1 \leq 3x_2 + 200 = 10(55 - x_2)$$

$$200 + 3x_2 = 550 - 10x_2$$

$$-200 + 10x_2 = -200 + 10x_2$$

$$\frac{13x_2}{13} = \frac{350}{13}$$

$$x_2 = 26.92307$$

hrs of class

hrs worked

$$x_1 = 55 - x_2 \rightarrow$$

$$x_1 = 28.076923$$

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New constraint: x_2 must be multiple of 5

Greatest x_2 val in our feasible region ≥ 25 :

There are multiple solns satisfying this, as

$x_2 \geq 25$ is bounded by $10x_1 - 3x_2 \geq 200$ &

$x_1 + x_2 \leq 55$; Plug in $x_2 \geq 25$ & solve,

we'll get $(27.5, 25)$ & $(30, 25)$.

So, one # hrs worked $27.5 \rightarrow 30$

satisfies all constraints.

-all these solns. maximise x_2 under constraints,

So Marko can choose soln. with 27.5

hrs worked or 30 worked if she wants

max. class time or more \$, respectively